Second midterm: decisions and loops

Information Analysis and Processing (TIA503). Duration: 75 minutes. Total: 100 points. October 22, 2026, from 16:00 to 17:15 (UTC-6).

Before you start

Open `VBA-Midterm.xlsx`, which contains the `Sales` and `Loan` worksheets with all starting data. Save the workbook as an Excel Macro-Enabled Workbook (`.xlsm`) before writing code.

Write two independent procedures in one standard module, with `Option Explicit` at the top. Use decisions, loops, variables, and worksheet operations covered in class. User-defined functions, arrays, UserForms, external libraries, and worksheet formulas that replace the requested VBA calculations are not allowed. Do not use `Select` or `Activate` to access cells.

Submit one ZIP file named with your full name, containing your `.xlsm` workbook and the exported `.bas` module. Keep both worksheet names and all input cells unchanged for the final submission. Allow 5 minutes to read, 30 minutes for Exercise 1 and 35 minutes for Exercise 2, and 5 minutes to check and submit.

Exercise 1. Sales commissions

This exercise is worth 50 points. The `Sales` worksheet has headers in row 1 and these records in columns A to C. All amounts are positive, all references are present, and each status is exactly `Paid` or `Pending`.

Ref	Amount (MXN)	Status
S-301	120,000.00	Paid
S-302	100,000.00	Pending
S-303	75,000.00	Paid
S-304	50,000.00	Pending
S-305	35,000.00	Paid
S-306	20,000.00	Paid
S-307	90,000.00	Pending
S-308	45,000.00	Paid

Write `Sub SalesCommissions()`. Find the last row containing a reference in column A, then process all records from row 2 with a `For` loop. The procedure must also work when another valid sale is added immediately below the existing records.

Use `Select Case` to assign the tier and commission rate from the sale amount:

Amount (MXN)	Tier	Rate
At least 100,000	High	4%
At least 50,000 and less than 100,000	Standard	2.5%
Less than 50,000	Basic	1%

Write the tier in column D and the commission (amount multiplied by rate) in column E. Use `If` to add commissions from `Paid` sales to commission payable and commissions from `Pending` sales to commission on hold. Count the sales in each status.

The summary labels are already in G1:G5. Write total sales amount in H1, commission payable in H2, commission on hold in H3, the number of paid sales in H4, and the number of pending sales in H5. Reset all accumulators before the loop so rerunning the procedure produces the same results.

Apply `$#,##0.00` to sale amounts, commissions, and H1:H3. Every cell reference inside the loop must use the row variable; do not write a separate calculation for each sale.

Exercise 2. A fixed monthly repayment

This exercise is worth 50 points. The `Loan` worksheet contains the following inputs, along with schedule headers in row 6 and summary labels in G1:G3.

Cell	Input	Value
B1	Opening balance (MXN)	60,000.00
B2	Monthly interest rate	1.5%
B3	Monthly payment (MXN)	7,000.00

Write `Sub LoanSchedule()`. Read all three inputs into variables. Assume a positive opening balance and payment, and a nonnegative monthly rate.

Before changing any report cell, check whether the monthly payment is greater than the first month's interest. If it is not, show `The payment does not cover the first month's interest.` and leave the procedure with `Exit Sub`.

After that check, clear the previous schedule in A7:E126 and the previous summary values in H1:H3. Use a `Do While` loop that continues while the balance is greater than zero and the month counter is less than 120.

For each month, starting in row 7, write five values in columns A to E: month number, opening balance, interest, payment, and closing balance. Calculate interest as opening balance multiplied by the monthly rate. The normal payment is the amount in B3. If it exceeds opening balance plus interest, reduce that month's payment to exactly the amount owed. Calculate closing balance as opening balance plus interest minus payment.

Increase the month counter, add that month's interest to the accumulated interest, and carry the closing balance into the next month. Use the row variable to write each schedule line. Keep full precision in calculations and use number formatting for the displayed amounts.

When the loop ends, write the number of months in H1, total interest in H2, and the last payment in H3. If a balance remains after 120 months, show `120-month limit reached.`. Apply `$#,##0.00` to the schedule amounts in columns B to E and to H2:H3.

Run the procedure with the supplied inputs. Then change B3 to 900 and check that the validation message appears without clearing the existing report. Restore B3 to 7000 and run the procedure again before submitting.